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Topic 81-160

A network of collagen fibers on which needlelike crystals of hydroxyapatite (a form of apatite, CaPO4) accumulate The Skeleton

Bone

Cartilage are the type of tissues which are progressively mineralized by:

Apatite

The first vertebrates had a_____skeleton

Cartilaginous

The_____appears in the early embryo as a strip of cells where the backbone will develop

Jawless Fishes

neural crest

in_____, commonest vertebrates were the conodont animals

Late Cambrian and Ordovician

_____Were jawless, generally armored

Ostracoderms

_____are the extinct genus of jawless fishes that lived in Devonian period.

Hemicyclaspis

They occur as phosphatic toothlike microfossils, termed elements.

Conodonts

Conodonts are divided into_____groups

3

_____more complex, with cones, bars and blades

Euconodonts

The three main morphotypes of conodonts are:

Coniform, Ramiform, Pectiniform

• The basal vertebrates, lacked jaws, jaws probably evolved during the_____

Ordovician

• Oldest jawbearing fishes were the_____

Placoderms

chondrichthyans were the first_____like cartilaginous fishes

Shark

Small fishes, mostly in the range_____ in length.

50–200 mm

The_____are Ancestors of most fishes today

Actinopterygians

The first good evidence of tetrapods were_____long

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The oldest fossil form, with reduced limbs, is_____in age

Jurassic

A part of amniotic egg that collects waste.

Allantois

_____are Distinguished by the pattern of openings in the side of the skull

Late Carboniferous

The three major skull patterns seen in amniotes:

anapsid, diapsid and synapsid

All modern amniotes produce the_____egg

Cleidoic

The oldest Turtles and relatives were_____eaters

small insect

First synapsids were Grouped loosely as "_____"

Pelycosaurs

Reptiles typically have _____bones in the lower jaw

Six

In mammals, there is a single bone in the lower jaw known as:

the dentary

As compared to the three tiny bones in human ear, reptiles have only one, known as:

stirrup or stapes

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Dinosaurs famously ruled the Earth for_____of the Mesozoic

160 Myr

The mammals achieved great diversity and abundance only after the extinction of:

Dinosaurs

A group, includes the dinosaurs, pterosaurs, crocodilians and birds.

Archosaurs

_____has a lightweight body, Narrow hatchet-shaped skull, a long narrow wing

Proficient flapping flyers

First true lizards are:

Lepidosaurs

The theropods and sauropodomorphs share the primitive_____pattern

reptilian hip

All other dinosaurs share a unique hip pattern in which the pubis has swung back and runs parallel to:

Ischium

fish-shaped animals, adapted to life in the sea

The ichthyosaurs

Marine reptile_____uses snake like neck for hunting the fish in water.

Plesiosaurs

The plesiosaur,_____Swam by beating their paddles in a kind of "flying" motion.

Cryptoclidus

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The most famous fossils of Oldest known bird were Found in Upper Jurassic sediments in southern Germany in

1861

Examples of Flightless ratites and ancestors of water birds

penguins and birds of prey

Small insect eaters in the Late Triassic and Early Jurassic Remained small through most of the:

Mesozoic

A classic example of an adaptive radiation due to development of teeth

Marsupials & Placentals

_____are early primates

Humans

• Homo sapiens means:

wise person

primates have Good_____vision

binocular

_____primates radiated in the Paleocene, Eocene and Oligocene

Basal

_____in Marmosets and spider monkeys are used as extra limbs in swinging through the trees.

Prehensile Tails

• The_____of Southeast Asia are the most primitive living apes

Gibbons

_____of the DNA of Gorillas, chimpanzees and humans is identical.

>94%

In_____, two French teams announced human fossils that were 6 Ma.

2001 and 2002

The famous skeleton of a female P. afarensis from Ethiopia, called_____by its discoverer

Lucy

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Lucy's brain is small, only_____for a height of 1–1.2 m

415 cm^3

Another human species, H . erectus had Brain size of_____in a body up to 1.6 m tall

830– 1100 cm^3

Homo erectus

Truly modern humans, H . sapiens, may have arisen as much as_____years ago

400,000

_____had large brains (on average, 1400 cm3)

Neandertals

_____, which concentrates on macroscopic (visible with the naked eye) plant remains

Paleobotany

Palynology is the study of_____.

fossil pollen and spores

Trace fossils are named on the basis of_____

shape and ornamentation

The study of trace fossils is often called _____

Ichnology

structures formed in hard substrates, such as limestone, shells or wood, for the purpose of protection

Borings

non-fecal pellets are regurgitation pellets of_____

birds and reptiles

Two models for the diversification of marine invertebrate life over the past 600 myr of good quality fossil records.

The Empirical Model, The Bias Simulation Model

the plateau appears clearly only in the_____data compilation.

family-level

A single equilibrium level is easier to understand in terms of a:

global equilibrium model

The Modern_____radiated dramatically over the last 100 myr

"fauna"

_____means change with improvement.

Progress

_____are observations of the appearance and disappearance of species.

Patterns

_____are hypotheses that seek to explain the patterns.

Processes

The replacement of brachiopods by bivalves is a famous example of:

Biotic replacements

First living organisms' bacteria:

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(3500 Ma)

Oldest multicellular eukaryotes: reported from rocks_____in Canada

1200 Ma

A process of reproduction in which a new individual arises by the fusion of two sex cells, or gametes, produced by different individuals

Sexual reproduction

Composed of many kinds of cells – muscle cells, nerve cells, epithelial cells and so on

Differentiated cells

the study of the distribution of species and ecosystems in geographic space and through geological time.

Biogeography

_____are ideal locations because they allow scientists to look at habitats that new invasive species have only recently colonized

Islands

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He noted many of the mammals of North America, such as bears, deer, squirrels, hedgehogs and moles, were found also in Eurasia • Travelled via Alaska, when climates were warmer

Georges Buffon

the founder of plant geography

German Alexander von Humboldt:

Augustin de Candolle of Geneva and Lamarck published a map showing France divided into _____regions with different ecological conditions.

five floristic

who was the first to classify the world's flora and show the results on maps?

Danish botanist Joakim Schouw

(113) Evolution – A Dangerous Idea!

French Revolution of_____led to a flowering of French science

1789

Power of the Church was broken:

18th century

_____of Natural History, became a powerhouse of ideas and debate in Europe.

New National Museum

In 1802,_____suggested that • 'Lower' organisms might also be found earlier in time and that they might gradually change into the 'higher' forms

Jean- Baptiste Lamarck

Geoffroy suggested evolutionary homologies and links between such widely different animals as:

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fish and cephalopods

Darwin had spent the _____ after his voyage in detailed research on many other areas of biology that provided evidence for evolution

40 years

_____is the central paradigm of the earth sciences and Theory of evolution is of the biological sciences

Plate tectonics

_____was the first to make a world map

Engler

The system of plant regions accepted today is very similar to that of:

Engler

"all species had been created within the area in which they are found today". • This was first formalized by:

British ornithologist Philip Sclater

Presented his theory of continental drift in (1912)

Pangaea Alfred Wegener

Croizat published his ideas in the 1950s and 1960s, his major presentation being his 1958 book:

Panbiogeography

Plants in High levels of heat and moisture

Megatherms

Plants in Moderate levels of heat and moisture

Mesotherms

plants in Low levels of heat and moisture

Microtherms

James Dana divided the_____into several different zones based on mean minimum temperature.

surface waters

_____in 1856 published Five depth zones and 25 faunal provinces along the coasts of the continents

Edward Forbes

three British zoologists Assembled_____of the distributions of fishes based on the patterns of distribution of 27 families.

30 maps

Sheer variety and volume of the works published by:

Alfred Wallace Candolle

New Guinea contributes only_____of the world's land area and some 10% of its species of terrestrial organism

3%

A species that exists as a series of subspecific forms is termed A_____Species

Polytypic

All chimpanzee subspecies lie to the north or east of this river

River Congo

A_____may be deterred from crossing an area of grassland; or a marsh organism may fail to travel across dry habitats to reach the next area of wetland.

forest species

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_____that leaves its forest- floor log, is exposed beetles, shrews and insectivorous birds.

Insect

_____Dive for their food and bring vegetable matter to the surface from greater depths.

Coots

The coots are messy eaters, and it is not difficult for gadwall to move in and collect some of the loot. This behavior is called

Klepto-parasitism

Patterns of Distribution

• Species may evolve in one region, spread to other locations and then become extinct in all but a restricted area where it survives. Species restricted in this way are said to be

Endemic

the two major factors influencing the degree of endemism in an area:

Isolation, Stability

Australia is indeed rich in endemics, many of which have a long geological history. This kind of 'fossil endemism' is called

Palaeo-endemism

Anything that tends to make it more difficult for a species to live, grow or reproduce in its environment may prove to be a:

limiting factor

Any regular change in physical or chemical conditions through space that creates a sequence of replacement of one species by another, among both animals and plants.

zonation

the predators can always turn to alternative food species if the numbers of their usual prey should be reduced by climatic or other influences. This is termed.

Prey Switching

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_____reduce the rate of growth of populations because of their negative influence on general fitness, survival and fecundity.

Parasites

The removal of certain species could create effects far in excess of what may originally have been expected. Influential species of this kind are known as.

Keystone Species

The loss of a_____can cause an avalanche of local extinctions.

Keystone Species

Microscopic plankton are capable of changing the depth at which they live, depending on conditions, and this can be regarded as a form of:

vertical migration

_____of a species can be measured by its geographical distribution

Success

The term_____is used for animals that play a particular role in the community, such as plant-sucking insects or insectivorous birds

guild

To cope with the physical demands of their environment

Species adapt

species tolerating one another, or becoming dependent on one another

Co- evolution

Pointing out that no two species have quite the same needs.

Individualistic concept

_____of certain species from an ecosystem would have little or no effect on the functioning of that ecosystem

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Removal

A possible alternative model is based on the supposition that all species are equally important to:

ecosystem function

certain species play key roles in the ecosystem, and when they are lost there is a sudden drop in the capacity of the ecosystem to function. This is sometimes referred to as:

The Rivet Hypothesis

_____is Defined in terms of inertia, or resistance to change.

Stability

Woody savannas – trees > 2 m high, cover only _____of the land surface

30–60%

Variations in climate also result from the pattern of movement of

air masses

The 'trade winds', found in both the Northern and Southern Hemispheres, meet in the region of the equator, and this is known as the:

intertropical convergence zone (ITCZ)

Approximately_____species of organism have been described

1.8 million

Of the bacteria, only about_____species have so far been described

4000

_____have always been most abundant in the low-latitude (tropical) regions.

Angiosperms

Greater extinction rates in the higher latitudes is the instability of climatic conditions over the past_____years

2 million

High-latitude organisms have_____than those from the low latitudes.

broader geographical ranges

General feature of biogeography first pointed out by_____in the 1970s

E.H. Rapoport

Katherine Smith and James Brown examined the diversity of_____as one proceeds deeper

fish

The Amazon basin contains approximately_____flowering plant species, whereas equivalent areas in Africa and in South-East Asia contain only about 40,000 each.

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90 000

All the land surface area of the hotspots together comprise only about_____of the Earth's terrestrial total

1.4%

The idea that continents could fragment and move across the face of the planet was first suggested by the German meteorologist_____in 1912

Alfred Wegener

The surface of the Earth, known as the lithosphere, is occupied by a number of areas known as:

tectonic plates

The_____move as well as the continents, the study of their movements is known as plate tectonics rather than continental drift

ocean floors

The fossil record suggests that all the early amphibian and reptile groups evolved in the continent called

Euramerica

_____are all the result of the break-up and dispersal of Gondwana

Southern continents

mainly the result of the appearance and disappearance of land or sea barriers within North America and Eurasia

Northern continents

Apart from humans and the aerial bats, only the _____spread naturally as far as Australia

rats

Human beings probably arrived_____years ago and brought the domestic dog (the ancestor of the dingo) about 3500 years ago

60 000 to 40 000

Geological studies show that_____is the smallest (about 1600 km2) surviving fragment of Gondwana

New Caledonia

New Zealand has Its total area,_____, makes it far larger than any other Pacific oceanic island

270 000 km

Effort by Naila Akram

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0.6 and 1 m

_____are 4000 species of mainly small animals that live in or close to water

Half-way land animals